

EEE4114F: Digital Signal Processing

Class Test

11 March 2024

SOLUTIONS

Please do not put your name or student number anywhere on this script.

Peoplesoft ID:

Information

- The test is closed-book.
 - This test has *four* questions, totalling 20 marks.
 - Answer *all* the questions.
 - You have 60 minutes.
 - An information sheet is attached.
-

1. (5 marks) A system is governed by the input-output relation

$$y[n] = \max_{-\infty < k \leq n} |x[k]|,$$

where $x[n]$ is the input and $y[n]$ is the output. The "max" operator returns the maximum value of the argument over the specified range, so in other words

$$\max_{-\infty < k \leq n} |x[k]| = \max\{\dots, |x[n-2]|, |x[n-1]|, |x[n]|\}.$$

- (a) Find the response $y_1[n]$ of the system to the input $x_1[n] = \delta[n]$.
- (b) Is the system linear? Why?
- (c) Is the system time invariant? Why?

- (a) The output of the system at a given instant is the maximum over all inputs up until (or to the left of, inclusive) that instant. The required output is

$$y_1[n] = \max_{-\infty < k \leq n} \delta[k],$$

which will be zero for $n < 0$ and one for $n \geq 0$. Thus the required response is $y_1[n] = u[n]$.

- (b) Suppose $x_2[n] = -\delta[n]$. The response will be $y_2[n] = u[n]$. If the system is linear then the response to $x_3[n] = x_1[n] + x_2[n] = \delta[n] - \delta[n] = 0$ should be $y_3[n] = y_1[n] + y_2[n] = 2u[n]$. However, the response is clearly zero, which is not equal to $2u[n]$. Thus the system is not linear.
- (c) Suppose $y_1[n]$ is the response of the system to the input $x_1[n]$. Then

$$y_1[n] = \max_{-\infty < k \leq n} |x_1[k]|.$$

Now suppose that $x_2[n] = x_1[n-c]$. The output will then be

$$y_2[n] = \max_{-\infty < k \leq n} |x_2[k]| = \max_{-\infty < k \leq n} |x_1[k-c]| = \max_{-\infty < m \leq n-c} |x_1[m]| = y_1[n-c].$$

Thus the system is time invariant.

2. (5 marks) Suppose the signal

$$x[n] = \sum_{k=-\infty}^{\infty} (-1)^k \delta[n - 2k]$$

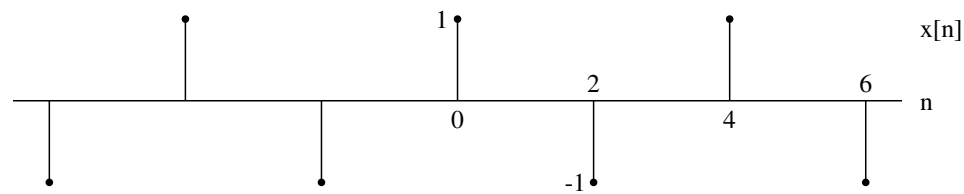
is provided as input to an LTI system with impulse response

$$h[n] = \delta[n] + \delta[n - 2].$$

(a) Sketch $x[n]$ over the range $-6 \leq n \leq 6$.

(b) Find and sketch the output $y[n]$.

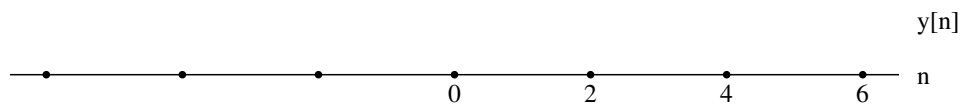
(a) Plot as follows:



(b) The output is

$$y[n] = h[n] * x[n] = (\delta[n] + \delta[n - 2]) * x[n] = x[n] + x[n - 2].$$

However, we observe that $x[n - 2] = -x[n]$, so $y[n] = 0$:



3. (5 marks) Answer the following:

(a) Find the DTFT of the following sequence:

$$f[n] = e^{-3n} \cos(3n)u[n].$$

(b) Find the inverse DTFT of the following:

$$X(e^{j\omega}) = \frac{e^{-j2\omega} - 1}{e^{-j2\omega} - 4}.$$

(c) (Bonus 2 marks) If $X(e^{j\omega}) = \text{DTFT}\{x[n]\}$ then Parseval's theorem states that

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_0^{2\pi} |X(e^{j\omega})|^2 d\omega.$$

Find the energy in the signal having DTFT

$$Y(e^{j\omega}) = \frac{1 - ae^{-j\omega}}{a - e^{-j\omega}}$$

where $-1 < a < 1$. Hint: the magnitude of the DTFT has a particularly simple form.

(a) We can write

$$f[n] = \frac{1}{2}e^{-3n}(e^{j3n} + e^{-j3n})u[n] = \frac{1}{2}(e^{-3+j3})^n u[n] + \frac{1}{2}(e^{-3-j3})^n u[n]$$

so

$$F(e^{j\omega}) = \frac{1}{2} \frac{1}{1 - e^{-3+j3}e^{-j\omega}} + \frac{1}{2} \frac{1}{1 - e^{-3-j3}e^{-j\omega}}.$$

(b) Since

$$\begin{aligned} X(e^{j\omega}) &= \frac{1 - e^{-j2\omega}}{4 - e^{-j2\omega}} = \frac{1}{4}(1 - e^{-j2\omega}) \frac{1}{1 - 1/4e^{-j2\omega}} \\ &= \frac{1}{4}(1 - e^{-j2\omega}) \left(\frac{1/2}{1 - 1/2e^{-j\omega}} + \frac{1/2}{1 + 1/2e^{-j\omega}} \right) \end{aligned}$$

we have

$$x[n] = \frac{1}{8}(1/2)^n u[n] + \frac{1}{8}(-1/2)^n u[n] - \frac{1}{8}(1/2)^{n-2} u[n-2] - \frac{1}{8}(-1/2)^{n-2} u[n-2].$$

(c) The squared magnitude of the transform is

$$|Y(e^{j\omega})|^2 = \frac{1 - ae^{-j\omega}}{a - e^{-j\omega}} \frac{1 - ae^{j\omega}}{a - e^{j\omega}} = \frac{1 - ae^{j\omega} - ae^{-j\omega} + a^2}{a^2 - ae^{j\omega} - ae^{-j\omega} + 1} = 1,$$

Thus the energy is

$$E = \sum_{n=-\infty}^{\infty} |y[n]|^2 = \frac{1}{2\pi} \int_0^{2\pi} |Y(e^{j\omega})|^2 d\omega = \frac{1}{2\pi} \int_0^{2\pi} d\omega = 1.$$

4. (5 marks) Consider the difference equation

$$6y[n] - y[n-1] - y[n-2] = x[n-1],$$

where $x[n]$ is the input and $y[n]$ is the output. What are the impulse response $h[n]$ and the frequency response $H(z)$ for the causal LTI system satisfying this difference equation?

The difference equation can be transformed to

$$(6 - z^{-1} - z^{-2})Y(z) = z^{-1}X(z),$$

or

$$6(1 - 1/2z^{-1})(1 + 1/3z^{-1}) = z^{-1}X(z).$$

The system function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{6(1 - 1/2z^{-1})(1 + 1/3z^{-1})} = \frac{z}{6(z - 1/2)(z + 1/3)},$$

which has poles at $z = 1/2$ and $z = -1/3$. To be causal the ROC must lie outside of the poles, so $|z| > 1/2$.

A partial fraction expansion on $H(z)$ can be obtained as

$$H(z) = \frac{1/5}{1 - 1/2z^{-1}} - \frac{1/5}{1 + 1/3z^{-1}},$$

so the impulse response is the inverse

$$h[n] = 1/5(1/2)^n u[n] - 1/5(-1/3)^n u[n].$$

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega$$

(CTFT)

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

(DTFT)

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz$$

(ZT)

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}}$$

(DFT)

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n + 1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$